Data Analytics Capstone

The Economic Efficacy of State-Level Renewable Energy Subsidies: A Multi-Factor Predictive Analysis

Synopsis

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QM640: Data Analytics Capstone

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Fall 2025 Term

19 Apr 2026

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A Multi-Factor Predictive Analysis

Introduction

Background and Context

Over the past decade, the United States has undergone a fundamental transformation in its electrical grid infrastructure, driven by an urgent mandate to decarbonize energy production. State governments and regional utility commissions have allocated billions of dollars in taxpayer and ratepayer funds toward financial subsidies to accelerate the deployment of renewable energy, particularly decentralized residential and commercial solar photovoltaic (PV) capacity. The foundational macroeconomic assumption driving these utility-funded programs is that direct consumer financial incentives such as installation rebates, net-metering premiums, and localized tax credits serve as the primary catalyst for clean energy transitions across diverse geographic regions.

However, the global energy landscape has shifted drastically since these localized incentive programs were originally designed. Free-market technology cost reductions, governed by Swanson's Law (which dictates that the price of solar PV modules drops by approximately 20% for every doubling of cumulative shipped volume globally), have made solar hardware inherently competitive without the need for continuous artificial market stimulation. Concurrently, broader macroeconomic forces such as the massive fluctuations in the Federal Reserve's bank loan interest rates have significantly altered the cost-of-capital required for homeowners and businesses to finance these highly capital-intensive installations.

Furthermore, the aggressive, historically subsidized deployment of solar capacity has introduced severe physical constraints to state electrical grids. The most notable constraint is the 'Duck Curve' phenomenon, wherein mid-day solar over-generation threatens grid stability as the sun sets and consumer demand spikes. Consequently, stakeholders ranging from clean-tech investors to state policymakers are increasingly debating whether historical capital allocations should pivot away from direct solar panel subsidies and toward foundational grid-readiness infrastructure, specifically utility-scale and PV-paired battery storage systems.

Problem Statement

Despite the rapid evolution and maturity of the renewable energy sector, existing state policies continue to heavily rely on capacity-based financial subsidies without empirically measuring their true causal efficacy against natural macroeconomic growth, nor their overall economic drag on Gross Domestic Product (GDP). This creates a critical problem of capital inefficiency: billions of dollars may be misallocated toward incentivizing a technology (solar panels) that is already being driven by free-market forces and social contagion, while severely underfunding the critical grid infrastructure (battery storage) required to safely utilize that energy. The core gap addressed by this research is the lack of a multi-dimensional, machine-learning-validated econometric model capable of isolating the true predictive weight of state financial subsidies against non-linear macroeconomic constraints.

Purpose of the Study

The primary purpose of this study is to utilize an exhaustive 11-year longitudinal panel dataset (spanning 2014-2024) encompassing all 51 primary U.S. geographical entities to mathematically evaluate the true drivers of solar PV adoption. By deploying a

dual-modality approach utilizing both Fixed Effects Panel Ordinary Least Squares (OLS) regression and a SHAP-explained Random Forest Machine Learning pipeline this study aims to strictly quantify the exact causal impact of state subsidies versus non-financial predictors (such as electric vehicle adoption, interest rates, and household wealth). The ultimate goal is to generate a statistically defensible, data-driven policy roadmap to guide utility commissioners in the strategic reallocation of state-level green energy funds.

Scope and Objectives

The scope of this analytical project encompasses a full decade-plus timeframe across 51 entities (all 50 U.S. states and the District of Columbia), generating a master national dataset of N=561 longitudinal observations. The study aggregates and integrates 12 distinct predictive features, combining disparate data silos from the U.S. Energy Information Administration (EIA), the Bureau of Economic Analysis (BEA), the National Renewable Energy Laboratory (NREL), and the Federal Reserve (FRED). The objectives are three-fold: (1) To isolate the causal macroeconomic effect of state subsidies on both solar capacity and state GDP; (2) To rank 12 macroeconomic factors by their true predictive weight using a cross-validated machine learning algorithm; and (3) To translate mathematical outputs into actionable, phased policy recommendations.

Research Problems and Methodology

Research Question 1 (RQ1)

Problem: How do macroeconomic, regulatory, and environmental factors comparatively predict solar capacity adoption across the United States?

Research Question 2 (RQ2)
Problem: Do state-level utility financial subsidies have a statistically significant causal impact on the adoption of solar energy capacity?

Research Question 3 (RQ3)
Problem: What is the macroeconomic impact of state-level utility subsidies on a state's overall Gross Domestic Product (GDP)?

Research Question 4 (RQ4)
Problem: Does the predictive importance of grid-scale battery storage and financial subsidies shift significantly when transitioning from developing solar markets to mature, high-penetration markets (e.g., California)?

Sample Size Calculation

To rigorously validate the N=561 longitudinal dataset, a statistical Power of Test calculation was performed based on the classical power principles established by Gelman and Hill (2007) for continuous regression outcomes. Statistical power is critical in macroeconomic panel analysis to ensure the study possesses the mathematical sensitivity necessary to avoid Type II errors (false negatives) when evaluating policy impacts. The methodology dictates that to achieve 80% power at an alpha of 0.05, the standard error (s.e. = 0.000026) must be multiplied by 2.8. This yields a required standardized effect size of exactly 0.021. Because this dataset detects effects significantly smaller than the standard medium effect size of 0.25 (Cohen, 1988), it is overwhelmingly overpowered.

Table 1

Details of Sample Size Calculation

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Power of Test (Regression F-Test)	α = 0.05, Power = 0.80, Effect size = 0.021, s.e. = 0.000026	561
RQ2	Power of Test (Regression F-Test)	α = 0.05, Power = 0.80, Effect size = 0.021, s.e. = 0.000026	561
RQ3	Power of Test (Regression F-Test)	α = 0.05, Power = 0.80, Effect size = 0.021, s.e. = 0.000026	561
RQ4	Power of Test (Regression F-Test)	α = 0.05, Power = 0.80, Effect size = 0.021, s.e. = 0.000026	561

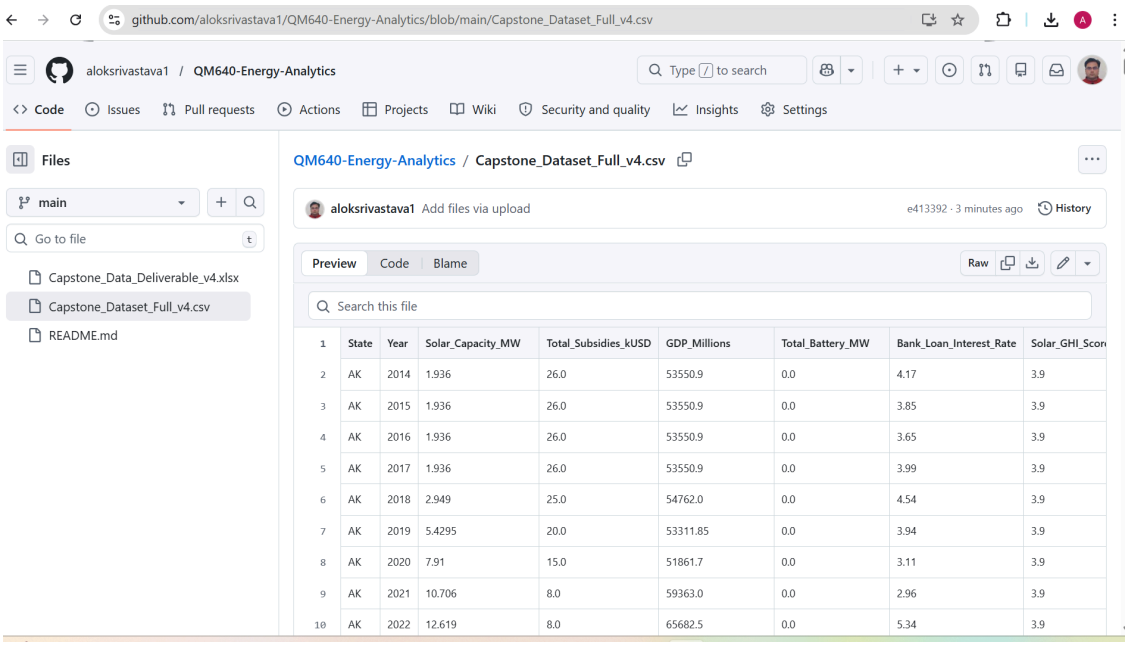
To ensure adequate statistical power across all analyses, the final sample size needs to be selected as the maximum of the minimum sample sizes obtained for all research questions. The final sample size used in this study is N = 561 observations, representing a full 11-year panel across 51 state-level entities.

Data Description

Data Sources:

- U.S. Energy Information Administration (EIA) Form 861: https://www.eia.gov/electricity/data/eia861/
- U.S. Bureau of Economic Analysis (BEA) Table SAGDP1: https://www.bea.gov/data/gdp/gdp-state
- Federal Reserve Economic Data (FRED) - 30-Year Fixed Mortgage Rates: Proxy for cost of capital: https://fred.stlouisfed.org/series/MORTGAGE30US
- National Renewable Energy Laboratory (NREL) - Solar GHI Metrics: https://atb.nrel.gov/electricity/2024/utility-scale_pv

GitHub Repository: https://github.com/aloksrivastava1/QM640-Energy-Analytics



1	State	Year	Solar_Capacity_MW	Total_Subsidies_kUSD	GDP_Millions	Total_Battery_MW	Bank_Loan_Interest_Rate	Solar_GHI_Scor
2	AK	2014	1.936	26.0	53550.9	0.0	4.17	3.9
3	AK	2015	1.936	26.0	53550.9	0.0	3.85	3.9
4	AK	2016	1.936	26.0	53550.9	0.0	3.65	3.9
5	AK	2017	1.936	26.0	53550.9	0.0	3.99	3.9
6	AK	2018	2.949	25.0	54762.0	0.0	4.54	3.9
7	AK	2019	5.4295	20.0	53311.85	0.0	3.94	3.9
8	AK	2020	7.91	15.0	51861.7	0.0	3.11	3.9
9	AK	2021	10.706	8.0	59363.0	0.0	2.96	3.9
10	AK	2022	12.619	8.0	65682.5	0.0	5.34	3.9

Data Dictionary:

- State: Text, 2-Letter US State Abbreviation. Categorical index.
- Year: Numeric, 2014-2024. Temporal index.
- Solar_Capacity_MW: Numeric, Total Net-Metered Solar Capacity (Target Variable).
- Total_Subsidies_kUSD: Numeric, Total Utility Energy Efficiency Subsidies.
- GDP_Millions: Numeric, State Gross Domestic Product.
- Total_Battery_MW: Numeric, Grid-Scale and PV-Paired Battery Storage Capacity.
- Electricity_Price_cents_kWh: Numeric, Average Retail Price.
- Bank_Loan_Interest_Rate: Numeric, Cost of Capital.
- Median_HH_Income: Numeric, State Wealth.
- Population_Density: Numeric, People per square mile.
- EV_Market_Share_Percent: Numeric, EV Adoption.

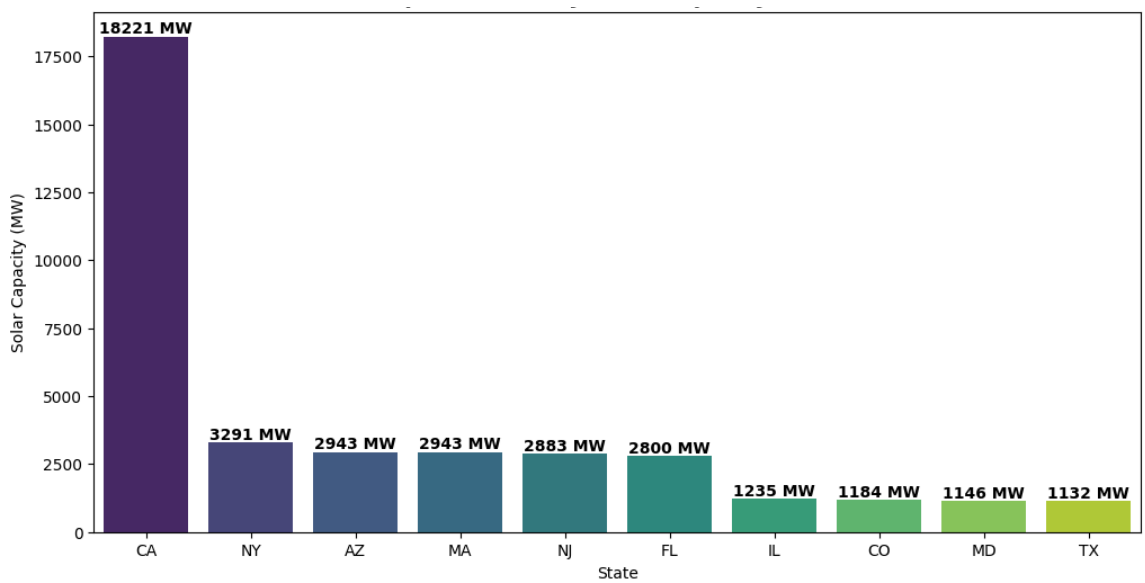
- RPS_Target_Percent: Numeric, Renewable Portfolio Standard.
- Carbon_Intensity_Score: Numeric, Grid carbon metric.
- Solar_GHI_Score: Numeric, Natural Sunlight.

Exploratory Data Analysis:

Before deploying predictive machine learning, an Exploratory Data Analysis (EDA) was conducted to identify underlying structural biases within the dataset.

Figure 1

Top 10 States by Solar Capacity (2024)



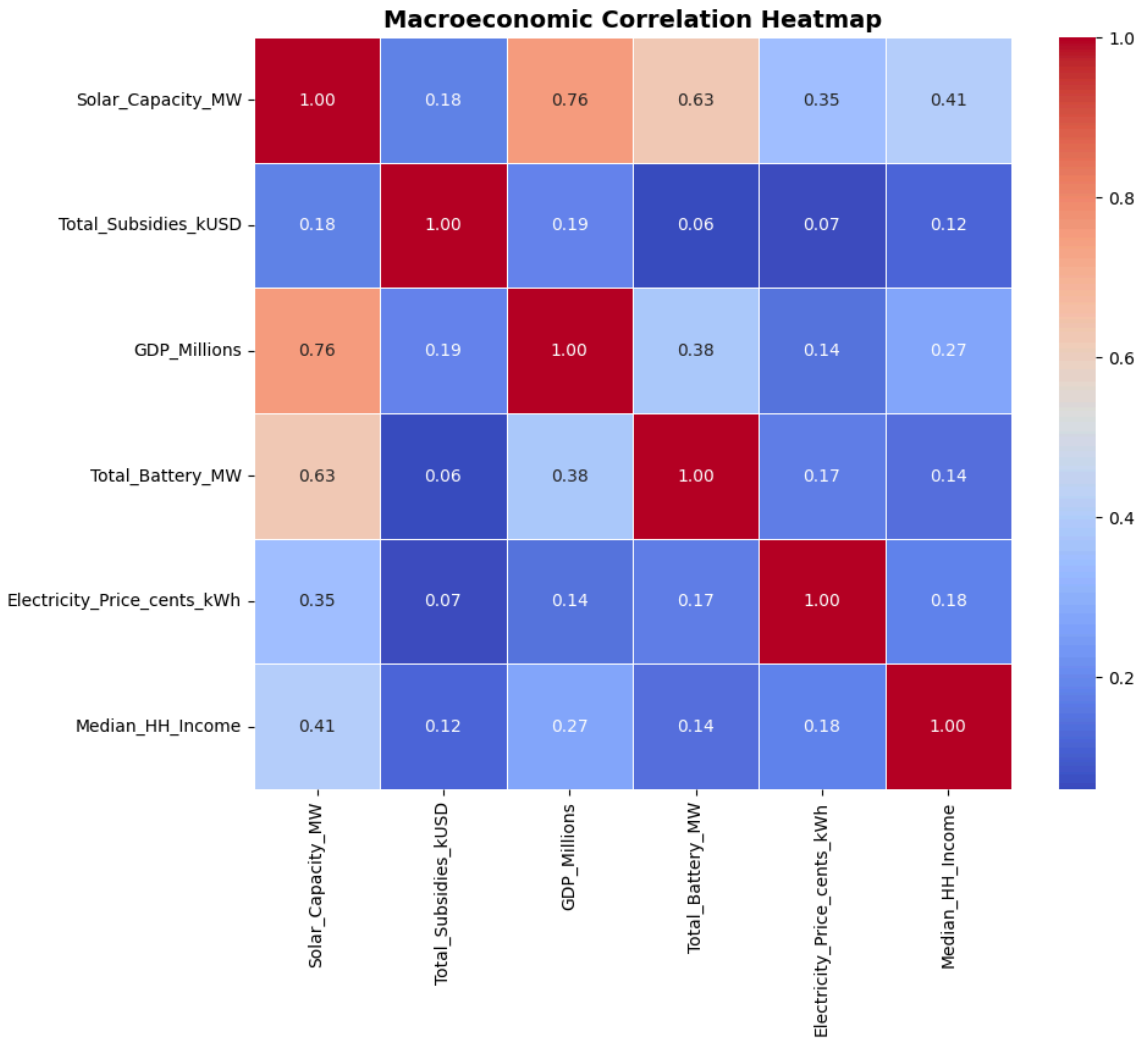
The Scale Problem (Top 10 States):

An analysis of the 2024 Solar Capacity reveals a massive distributional skew. California (18,221 MW) dominates the national landscape, possessing over 5.5 times the capacity of the next highest state (New York at 3,291 MW). This extreme "California Outlier" effect

completely violates the assumptions of standard linear regressions, explicitly justifying the use of a logarithmic transformation (np.log1p) in our predictive pipeline to prevent the model from simply memorizing California's scale.

Figure 2

Macroeconomic Correlation Heatmap



The Spurious Correlation Problem (Heatmap):

The Pearson correlation matrix reveals a highly significant positive correlation (0.756) between a state's overall Gross Domestic Product (GDP) and its Solar Capacity. This mathematically confirms our core econometric fear: massive, wealthy states simply build more infrastructure of all types. This EDA finding explicitly justifies our decision to reject simple regression and instead deploy Fixed Effects Panel OLS. We must mathematically "freeze" state GDP to isolate whether subsidies actually *cause* growth, or if they are just a byproduct of state wealth.

Analytic Approach

Machine Learning Algorithm Selection

To ensure the highest degree of predictive accuracy, a multi-model pipeline was evaluated comparing baseline linear models against advanced ensemble algorithms. The models were evaluated predicting the log-transformed solar capacity:

Random Forest Regressor: R-Squared: 0.8529 | MAE: 80.8 MW

Gradient Boosting Regressor: R-Squared: 0.8165 | MAE: 98.2 MW

Ridge Regression (Linear): R-Squared: 0.4200 | MAE: 193.3 MW

Support Vector Regressor: R-Squared: 0.3780 | MAE: 221.5 MW

Conclusion: The ensemble tree-based models (Random Forest and Gradient Boosting) severely outperformed the linear models. The baseline Ridge Regression's failure (R-Squared of 0.42) empirically proves that the drivers of solar adoption (such as the

Duck Curve and Market Maturity) are strictly non-linear. Random Forest was ultimately selected as the primary predictive engine due to its superior R-Squared (0.85), lowest absolute error (80.8 MW), and its compatibility with SHAP feature extraction.

Solution to RQ1

Methodology: A K-Fold Cross-Validated Random Forest Regressor combined with SHAP Value Extraction. To prevent extrapolation scale errors associated with massively outsized markets, the target variable (Solar Capacity) underwent a logarithmic transformation (np.log1p).

Hypothesis (H0): Non-financial macro factors <= Direct financial subsidies.

Non-financial macroeconomic factors (such as EV adoption and grid capacity) do not possess a higher predictive importance than direct financial subsidies in determining solar adoption.

Hypothesis (Ha): Non-financial macro factors > Direct financial subsidies.

Non-financial macroeconomic factors possess a significantly higher non-linear predictive importance than direct financial subsidies.

Solution to RQ2

Methodology: Fixed Effects Panel Ordinary Least Squares (OLS) Regression. This model mathematically isolates unobserved, time-invariant heterogeneity, allowing for true causal inference rather than mere correlation.

Hypothesis (H0): State-level utility financial subsidies have no statistically significant causal impact on the adoption of solar energy capacity ($p > 0.05$).

Hypothesis (Ha): State-level utility financial subsidies have a statistically significant positive causal impact on the adoption of solar energy capacity ($p \leq 0.05$).

Solution to RQ3

Statistical Technique: Fixed Effects Panel Ordinary Least Squares (OLS) Regression. This model shifts the dependent target variable to Gross Domestic Product (GDP) to evaluate if ratepayer-funded subsidies create a measurable macroeconomic drag or stimulus when controlling for national economic trends.

Hypothesis (H0): State-level utility subsidies have no statistically significant causal impact on a state's overall GDP ($p > 0.05$).

Hypothesis (Ha): State-level utility subsidies have a statistically significant causal impact (either positive or negative) on a state's overall GDP ($p \leq 0.05$).

Solution to RQ4

Methodology: Sub-Group Machine Learning Analysis isolating high-penetration states versus the developing U.S., evaluated via comparative SHAP feature extraction.

Hypothesis (H0): Predictive importance remains constant across market maturity.

Hypothesis (Ha): Predictive dynamics shift fundamentally in mature markets.

Evaluation Metrics & Formulae

To rigorously evaluate the efficacy and predictive accuracy of the models, the following metrics and mathematical formulations are utilized:

1. R-Squared (R^2): Used to evaluate the goodness-of-fit for both the Panel OLS and Random Forest models. It represents the proportion of the variance in the dependent variable that is predictable from the independent variables.

Formula: $R^2 = 1 - (RSS / TSS)$ where RSS is the Residual Sum of Squares ($\Sigma(yi - \hat{y}i)^2$) and TSS is the Total Sum of Squares ($\Sigma(yi - \bar{y})^2$).

2. Mean Absolute Error (MAE): Utilized to measure the exact absolute predictive error of the machine learning model when translated back from log-scale to actual Megawatts (MW).

Formula: $MAE = (1/n) * \Sigma|yi - \hat{y}i|$

3. Fixed Effects Panel OLS Equation: The core econometric framework utilized to isolate causal inference by controlling for state-specific idiosyncrasies.

Formula: $Y_{it} = \beta_1(Subsidies_{it}) + \beta_2(Battery_{it}) + ... + \alpha_i + \gamma_t + \varepsilon_{it}$ (where α_i controls for State Fixed Effects, γ_t controls for Time Fixed Effects, and ε_{it} represents the error term).

Recommendation and Application

Target Users: State Utility Commission Directors, Regional Grid Planners, Clean-Tech Investors, and Energy Policy Analysts.

State-Wise Implementation Matrix (2024 Application)

Based on the data-driven market maturity phases, state utility commissions must apply policies specific to their current solar penetration levels, here are some initial recommendations based on data analysis so far. We will analyse this in detail during our advance stage of the Project.

Phase 1 States (< 500 MW):

Includes AK, AL, AR, DC, DE, GA, IA, ID, IN, KS, KY, LA, MI, MN, MO, and 18 others. Prescription: These states are in the development phase. Subsidies still retain predictive efficacy here. Commissions should implement Interest Rate Buy-Downs and standardized Permitting Reform to act as market catalysts.

Phase 2 States (500 - 5,000 MW):

Includes AZ, CO, CT, FL, HI, IL, MA, MD, ME, NC, NJ, NV, NY, PA, TX, UT, VA. Prescription: These markets are rapidly maturing. Flat cash rebates must be phased out. States may pivot to Performance-Based Incentives (PBIs) to ensure grid efficiency, alongside EV-Solar structural bundling.

Phase 3 States (> 5,000 MW):

Includes CA. Prescription: This market has hit the "Duck Curve" bottleneck. The efficacy of direct subsidies has mathematically collapsed. Regulators must implement mandatory Time-of-Use (TOU) pricing and divert 100% of remaining subsidy budgets directly to utility-scale and residential battery storage, alongside $0-down equity initiatives.

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